Build your own

Robi

Pack 05

Stages 15-18:
Begin building
Robi's left arm

DeAGOSTINI
MODEL SPACE

Build your own Robi

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DeAGOSTINI MODEL SPACE™

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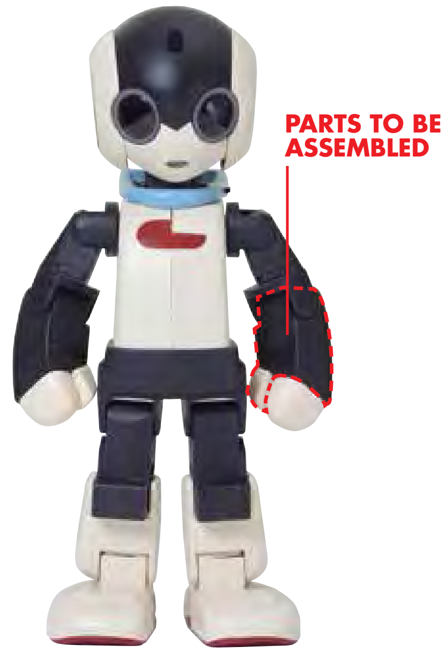
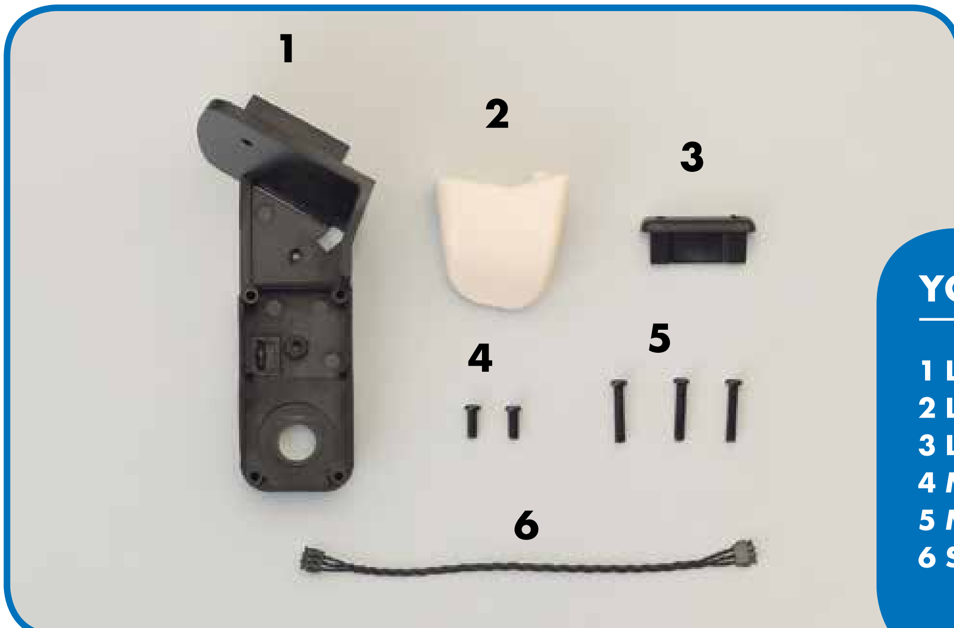
STAGE 15: ASSEMBLING ROBI'S LEFT FOREARM

In this stage you will start assembling Robi's left arm, beginning by fitting the thumb to the forearm frame. You will then prepare another servo cable by fitting it with protective pads.

Robi's left arm is symmetrical to his right, so the assembly process is nearly identical. Again, the servos that operate his arm's movement will be fitted onto a framework, the forearm section of which is supplied with this stage. After starting the physical assembly, you will go on to prepare the servo cable that will carry power to Robi's forearm servo on completion. As always, care must be taken when handling delicate electronic parts.



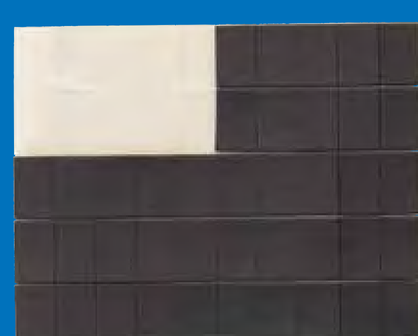


YOUR PARTS

1 Left forearm frame
2 Left thumb
3 Left finger mount
4 M2 x 5mm pan-head screws (x 2)
5 M2 x 10mm pan-head screws (x 3)
6 Servo cable (70mm long)

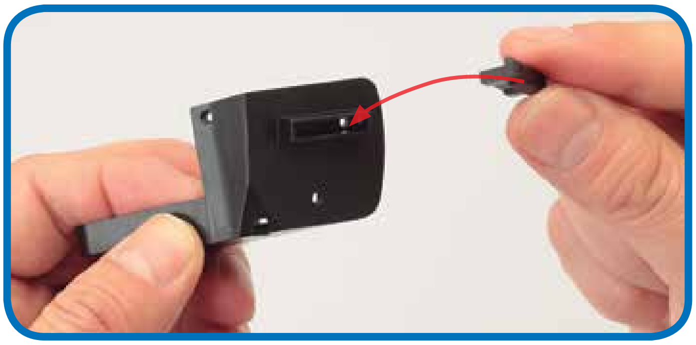
SAVED PARTS

You will be using the protective pads supplied with Stage 3, so be sure to have these to hand before beginning.



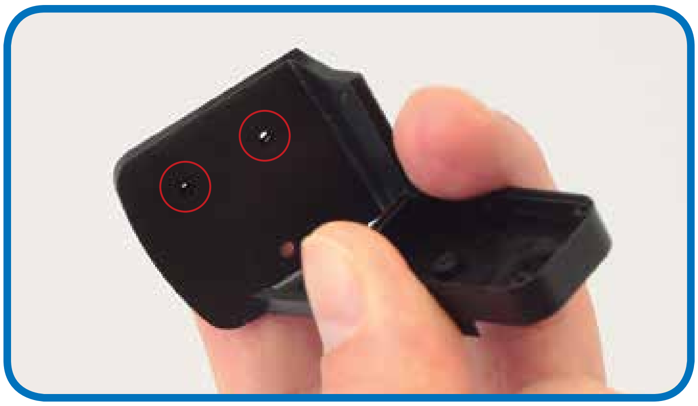
Protective pads (Stage 3)

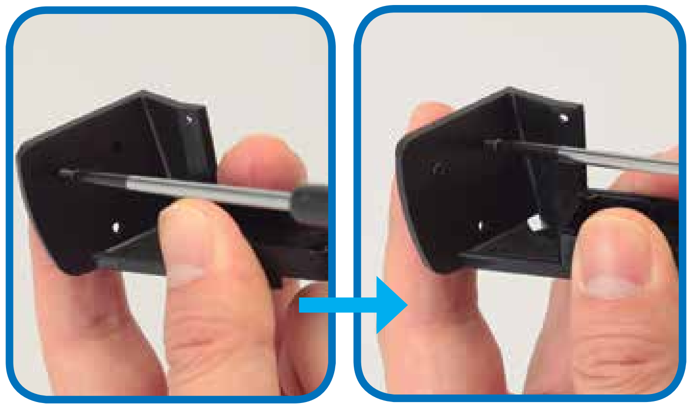
THE FINGER MOUNT



1 Align the left forearm frame and finger mount as shown. The mount can be fitted either way up.

2 Press the tongue of the finger mount into the long slot in the end of the forearm frame.

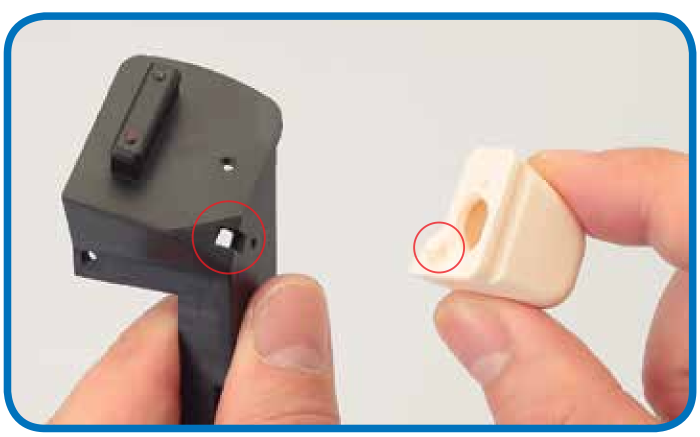


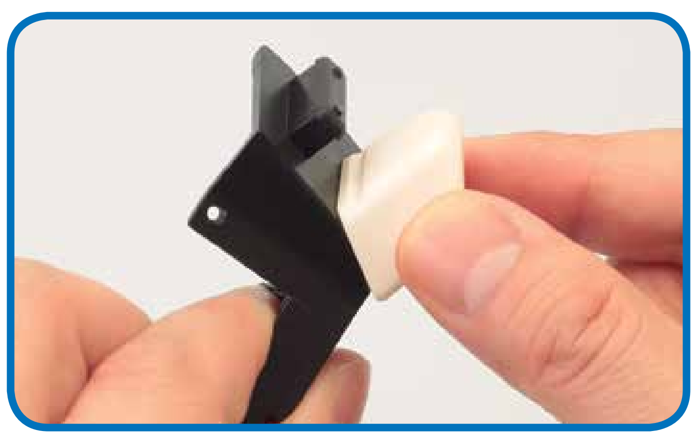


3 Turn the forearm frame over to check that the holes in the finger mount are in line with the two screw holes in the forearm frame.

4 Fit two M2 x 10mm pan-head screws through the holes and carefully tighten them to secure the finger mount.

THE THUMB





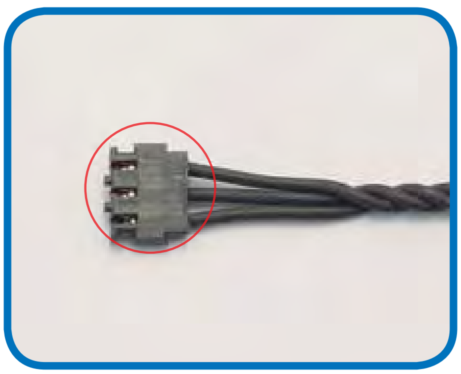
5 Line up the thumb and the forearm frame, noting the position of the square tab on the thumb and the matching hole in the frame, circled in red.

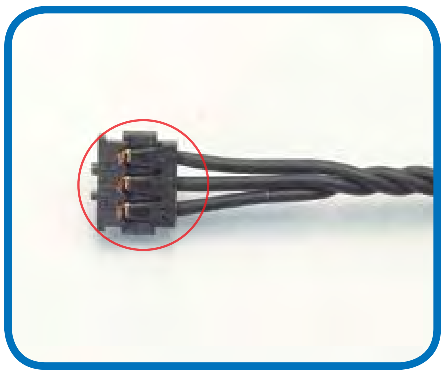
6 Press the parts together so that the tab locks into the hole and the thumb fits flush against the frame.

7 Turn the forearm frame over and check that the hole in the thumb is in line with the remaining screw hole in the forearm frame.

8 Fit an M2 x 5mm pan-head screw in the hole and carefully tighten it, to secure the mount.

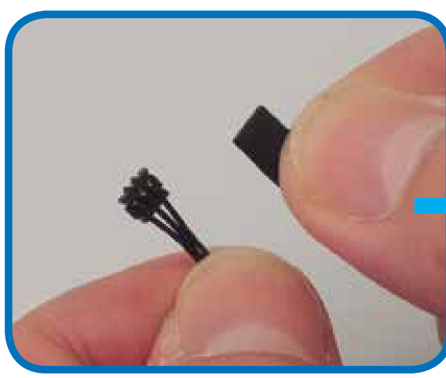
THE SERVO CABLE





9 Next, prepare the servo cable supplied with this stage. As you have done with previous cables, identify the side of the connector with PUSH printed on it. The protective pad will be fitted to this side.

10 Remove a protective pad from the sheet saved from Stage 3, and press it into place as shown onto the servo cable's connector. Repeat for the connector at the other end of the cable.



Assembled left forearm frame

The first parts of Robi's left arm have been assembled, and another servo cable is ready for use in the coming assemblies.

STAGE 16: SET THE ID NUMBER OF ROBI'S LEFT FOREARM SERVO

In this stage, you will set the ID number of the servo that acts as an elbow joint to move Robi's left forearm, before fitting the part onto the left forearm frame, ready for use.

The work in this assembly will be similar to that of Stage 8, when you wired and set the ID number of the corresponding servo for Robi's right arm. Even so, make sure you follow the assembly steps closely, as it can be difficult and, for the electronic parts such as the servo cables, even damaging, to re-open the assembly should you need to go back and perform the steps again.

Similarly, when setting the servo's ID, be absolutely sure you choose the correct number, as Robi may not work properly if the part is set to the wrong number when operated.

PARTS TO BE ASSEMBLED

YOUR PARTS

1 Servo motor (for left elbow)

SAVED PARTS

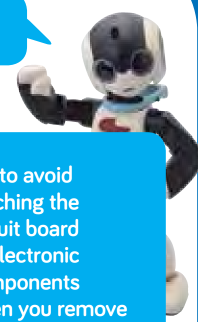
You will be using parts from the previous stages, so be sure to have these to hand before beginning.

Servo cable and left forearm frame (Stage 15)

Neck stand (Stage 8)

CONNECTING THE SERVO CABLE

BE CAREFUL!



Try to avoid touching the circuit board or electronic components when you remove the cover from the servo.

1 Use a screwdriver to undo the four screws securing the cover of the servo, and take each one out.

2 Carefully lift the cover away from the other side of the servo.

3 Take either end of the connector cable saved from the previous stage and carefully fit it in one of the two sockets circled in red. It does not matter which one. Gently push it until it locks in place.

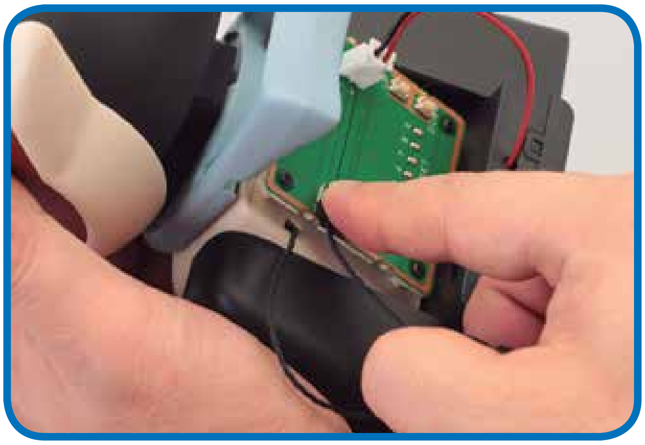
TIP!

If you chose not to fit the protective pad in Stage 15, you can fit it now, which will help to secure the connector in place.

4 Carefully feed the free end of the cable up through the circular hole in the servo cover. Make sure that the cover is the right way up.

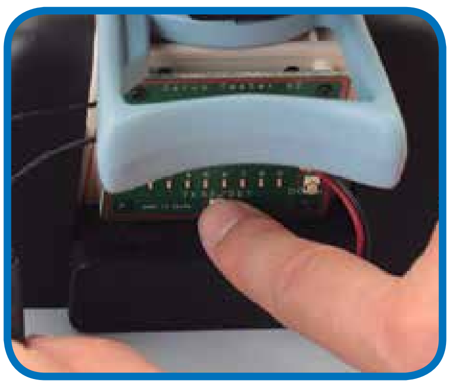
5 Fit the cover onto the servo casing, then turn the servo over and insert two of the screws you removed in Step 1. Tighten them lightly, as you will be removing them again in this assembly.

TESTING THE SERVO

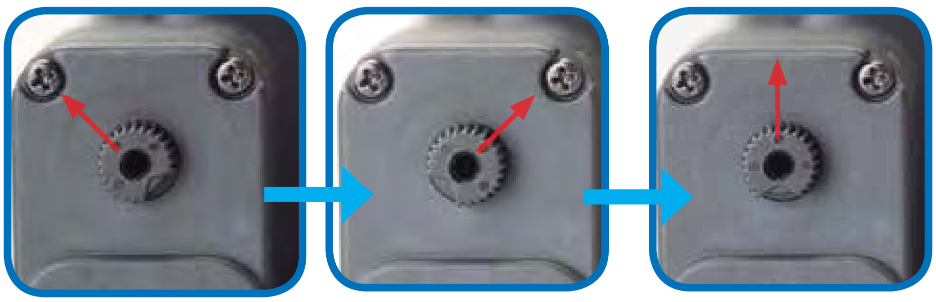


6 Take the neck stand and unplug the black servo cable from the socket on the left of the test board. Plug the cable leading to this stage's servo in its place.

7 Turn the test board to ON. All the LEDs should blink twice, then turn off to leave only number 1 lit.

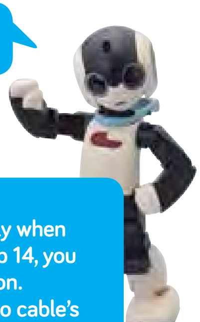


8 Now press the TEST/SET switch briefly to run the servo test.

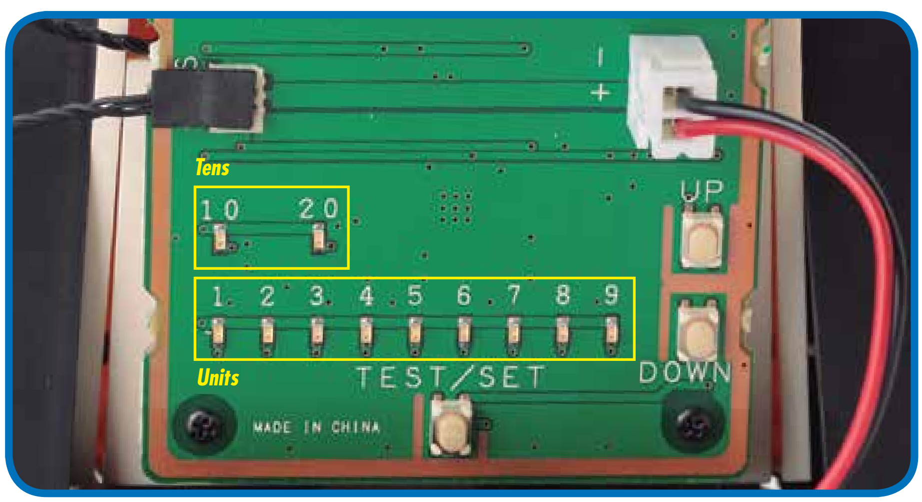


9 The shaft at the top of the servo casing should rotate 45 degrees to the left, stop, then move back across to 45 degrees to the right. After you have run the test, turn off the power switch on the battery pack.

HELP!

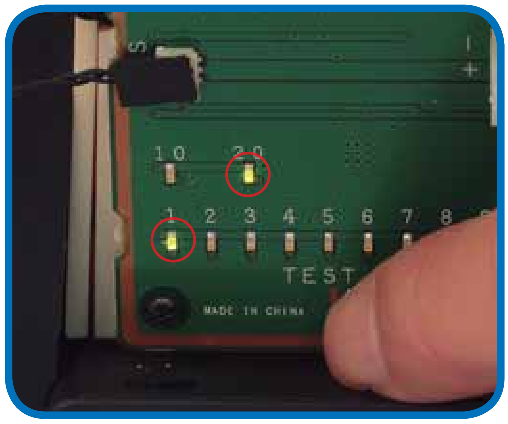


If the LEDs flash rapidly when you run the test at Step 14, you have a faulty connection. Double-check the servo cable's connection to the socket on the test board, TURNING THE POWER PACK OFF BEFORE YOU DO. If the light still flashes, carefully unscrew the servo housing and check that the internal connector plug is in place. If you pressed the wrong button, reset the board by holding down the TEST/SET switch for several seconds.



10 Re-familiarise yourself with the lights and switches on the test board, which you have used to set the servos' ID numbers. Each time you press the UP switch, the ID number increases by one, and the result is displayed by the two banks of LED lights. The top row reads in tens and the bottom in units. For example, to set the ID to 21, press UP 20 times. This will light up the 20 LED and the 1 LED. (If you overshoot, press the DOWN switch to take it down one step at a time.) When the ID number is correct, you set it by pressing and holding the TEST/SET switch.

SETTING THE ID NUMBER

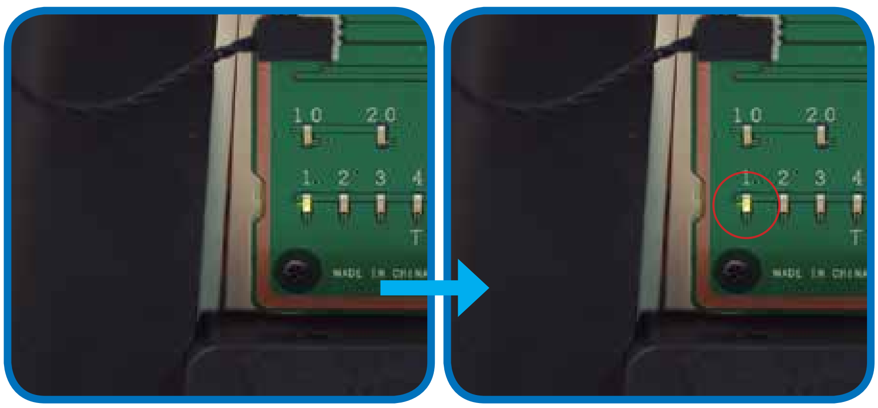


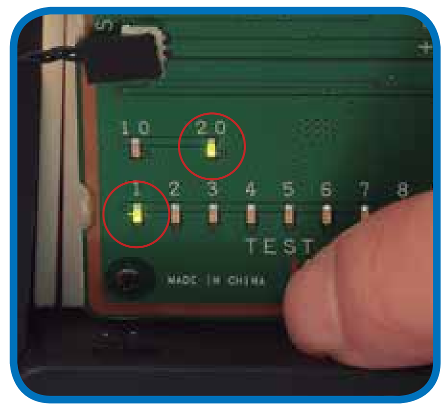
ID=21

11 To set the ID number of the servo to 21, press the UP button 20 times. The 20 and 1 LEDs should remain lit once you have done this.

12 Confirm the setting by pressing the TEST/SET button. The 20 and 1 lights should flash, then go out.

CHECKING THE ID NUMBER



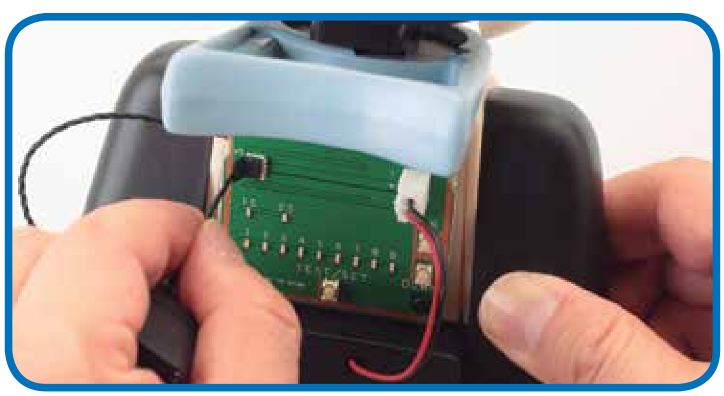


13 It is vital that each of Robi's servos is programmed to the correct ID number, or his operation may malfunction on completion. To begin checking the ID number, leave the servo connected, switch the power OFF and then back ON. Only the 1 LED should light up.

14 To check the ID number, press TEST/SET. Now the 20 and 1 LEDs should light up.

TIP!

It is normal for the 1 LED to light up when you switch on. It will only change to give you the ID number of the servo when you press TEST/SET. If you made a mistake and set the wrong ID number, you can reset it by going through the setting procedure again.

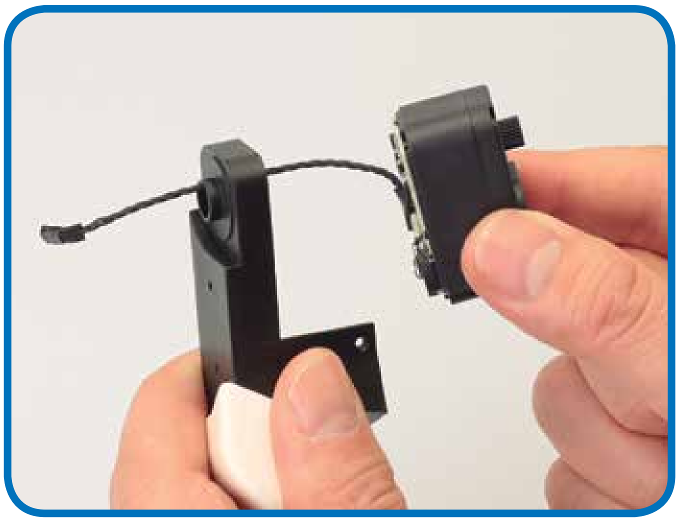


15 Once all is well, switch the power pack OFF, then carefully unplug the servo from the test board. It will retain its setting, ready to be fitted to Robi's left arm.

ATTACHING THE SERVO

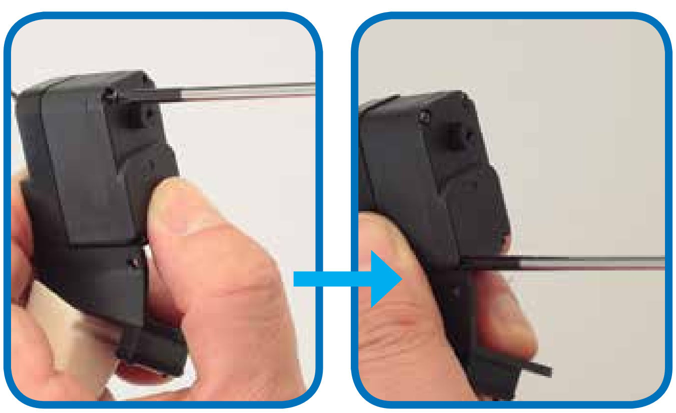
16 Remove the two screws fitted temporarily in Step 5 and discard the cover as you will not be using it again. Keep the screws.

17 Now line up the open servo with the left forearm frame. The servo cable will fit through the circular hole at the top of the frame (circled).



18 Carefully feed the servo cable through the hole in the forearm frame.

19 Press the servo into place at the top of the forearm frame. Make sure the servo cable is not bent or caught in any way as you do this.



20 Using the four screws you removed in Step 1, secure the servo to the forearm frame.

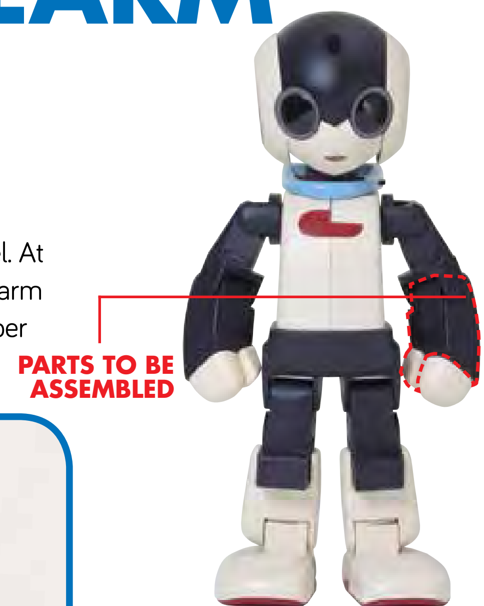
Assembled forearm

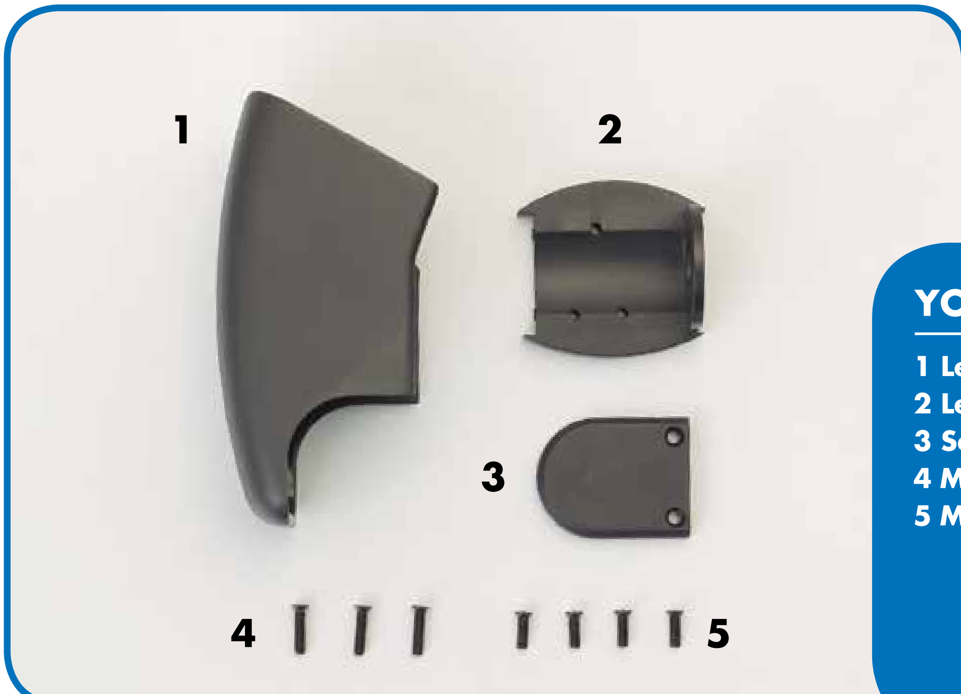
The next of Robi's servos is in place, ready to be joined to the rest of his body in future stages.

STAGE 17: COMPLETING ROBI'S LEFT FOREARM

In this stage, you finalise the parts that make up Robi's left forearm, from his elbow joint and forearm cover down to his hand.

In this session, you continue to assemble Robi's left forearm, carrying on from setting the ID number of the forearm servo in the last stage. This time, you will fit the elbow joint and forearm cover, before closing the assembly with the servo front panel. At the end of this session, the left forearm will be ready to be joined to the upper arm servo in coming stages.





YOUR PARTS

1 Left forearm cover
2 Left elbow joint
3 Servo front panel
4 M2 x 8mm countersunk screws x 3
5 M2 x 6mm countersunk screws x 4

SAVED PARTS

You will be using parts from the previous stages, so be sure to have these to hand before beginning.

Left forearm assembly (Stage 16)

THE D-CUT

The round shafts of Robi's servos are moulded with a small flat edge on one side – seen end-on, they are shaped like the letter 'D' – that fits into a correspondingly shaped recess on the part that the servo will drive. This ensures that the parts can be set correctly when in the neutral position.

A 'D-cut' shaft

Left photo: The servo's shaft is termed 'D-cut' because, viewed end-on, it isn't completely circular but has a D-shaped outline. Right photo: The hole into which the D-cut servo shaft will fit has a similarly shaped outline.

Neutral position

The advantage of using a D-cut shaft and joint set-up is that it makes it very easy to assemble Robi's servos and their accompanying parts correctly. When a servo and a part are aligned correctly, as in the far left photo, they will fit together easily, but if they are not lined up as they should be, they will struggle to fit together and result in an off-centred neutral position (near left). If this happens, do not force the part. Instead, realign it with the servo shaft until it fits easily.

Warning: applying too much force when fitting a part over the servo's shaft can result in damage to both items!

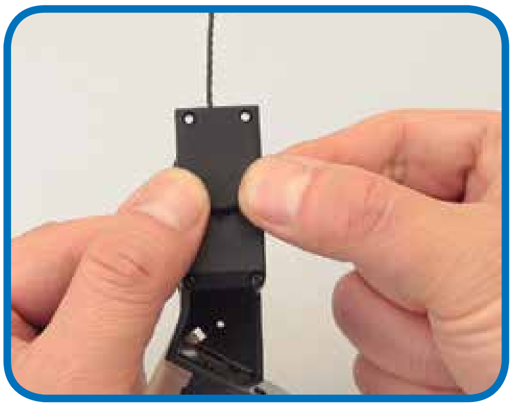
Restoring to the neutral position

If something goes wrong when you are testing one of Robi's servos – for example, if the power is accidentally disconnected and the servo's shaft stops turning and remains at an angle – you will need to reset the servo to its neutral position. To do this, identify the flat on the servo shaft (left photo) and place the part it will operate over it at the same angle (right photo). Then, very gently, rotate the part back to the neutral position, with the servo shaft's flat facing downwards.

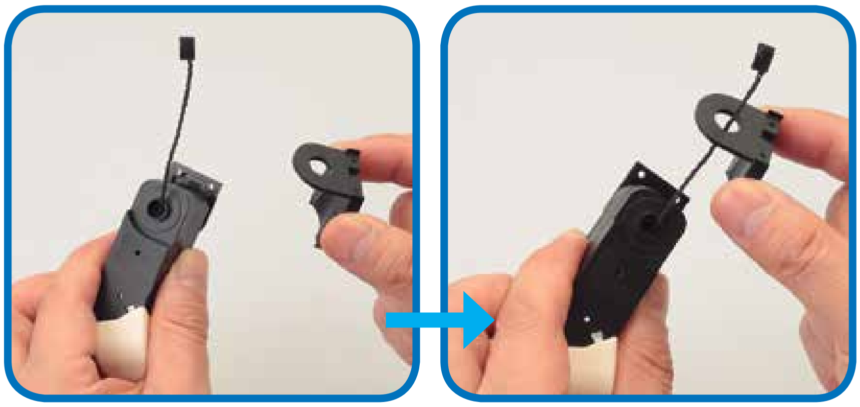
COMPLETING THE LEFT FOREARM

1 Take the left forearm assembly and the servo front panel supplied with this stage, and line up the D-shaped servo shaft with the correspondingly shaped socket on the underside of the servo cover. Be absolutely sure that the parts are set correctly together to avoid damage to either part – for more information, see the box about D-cuts on the previous page.

2 Join the parts so that they fit snugly together. Do not force the parts as you combine them.



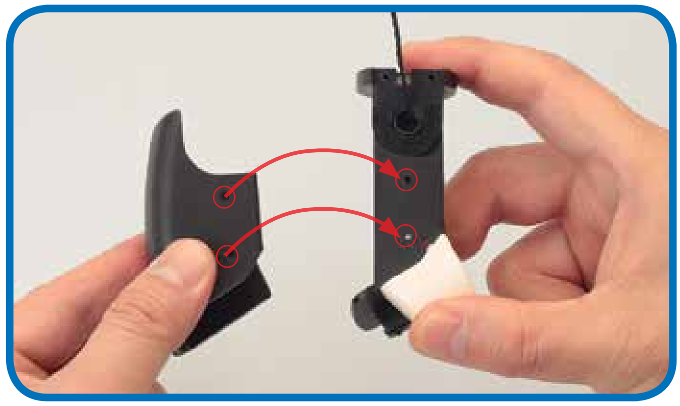
3 When you are certain that the parts are set straight to one another, press evenly on the panel to secure it to the servo shaft.



4 Turn the assembly around so that the side from which the servo cable emerges is facing you. Then, with the left elbow joint positioned as shown, lower the part over the cable, which should run through the hole in its centre.

5 Run the left elbow joint along the whole of the servo cable, then fit it onto the top of the left elbow frame, as shown.

6 The top edge of the elbow joint is moulded to accommodate the upper tip of the servo cover fitted in Step 2, so gently press these together.



7 Hold the parts together, then insert and tighten two of the longer, M2 x 8mm countersunk screws into the two holes shown above.

8 Now prepare to add the left forearm cover to the assembly, lining up the holes circled above.

9 Join the parts, making sure that the circled holes are properly lined up with one another, ready to take the two M2 x 6mm countersunk screws.

10 Insert and tighten both M2 x 6mm countersunk screws into the holes to secure the forearm cover.

11 Finally, locate the circled hole on the left forearm frame, and insert and tighten another M2 x 6mm countersunk screw into it.

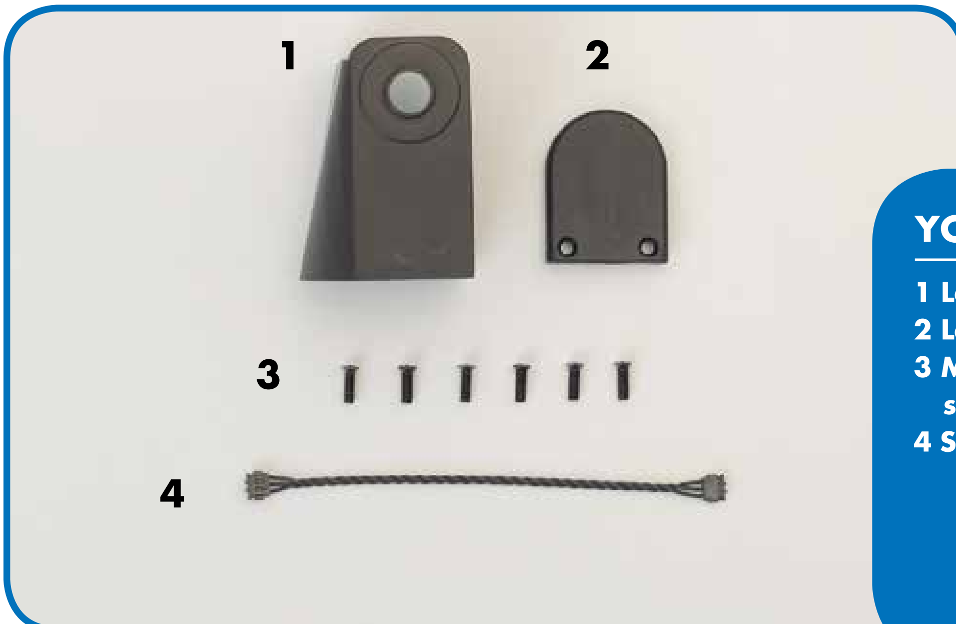
Assembled forearm

Robi's left forearm is now complete, and ready to be joined to his upper arm in future stages.

STAGE 18: BUILDING UP ROBI'S LEFT UPPER ARM

In this stage, you start building up Robi's left upper arm, beginning by sealing in his elbow before fixing the left upper arm frame itself.

The upper arm frame will join to the forearm assembly at the elbow, which you will seal off from the rear with the elbow back panel also supplied with this stage. As you do this, you will feed the forearm's servo cable through a small trough set into the side of the back panel and frame. It is very important that the cable fits through this recess freely, without getting trapped or caught between either part, as this may damage the cable and later affect the operation of Robi's arm. Lastly, you will prepare this stage's servo cable for safe use, as you have done in previous stages.



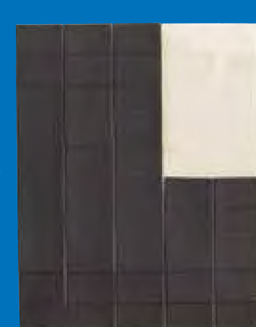
YOUR PARTS

1 Left upper arm frame
2 Left elbow back panel
3 M2 x 6mm countersunk screws x 6
4 Servo cable (70mm)

SAVED PARTS

You will be using parts from the previous stages, so be sure to have these to hand before beginning.

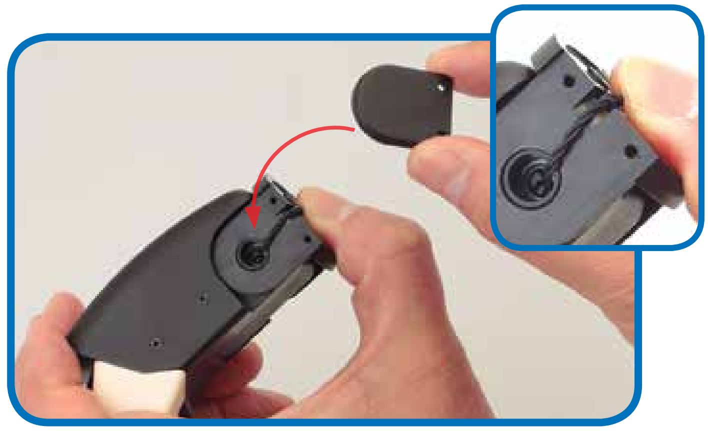
Left forearm assembly (Stage 17)



Protective pads (Stage 3)

FITTING THE LEFT ELBOW BACK PANEL

1 Take the left forearm assembly (built over previous stages). Fold the servo cable back and into the recess at the elbow, as shown.



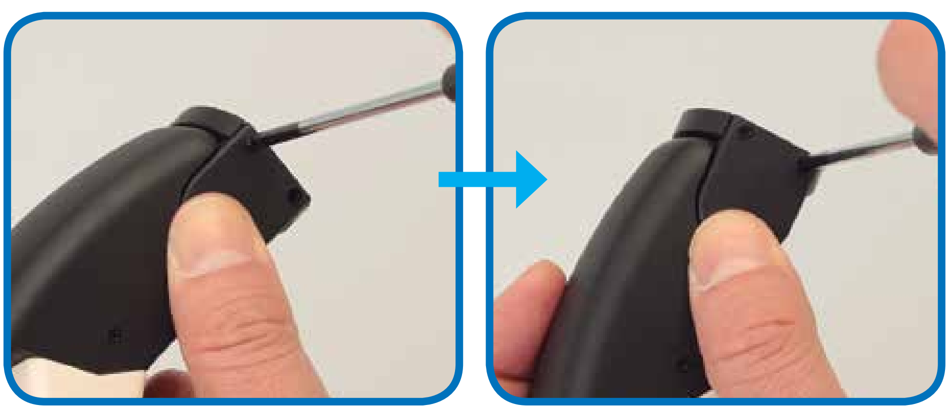
2 Holding the cable in the recess, place the left elbow panel onto the correspondingly shaped space on the forearm frame.

3 Press the elbow panel into position, so that its curved edge fits into the frame neatly.

4 Press the part firmly into place, making sure the cable isn't trapped in any way.

BE CAREFUL!

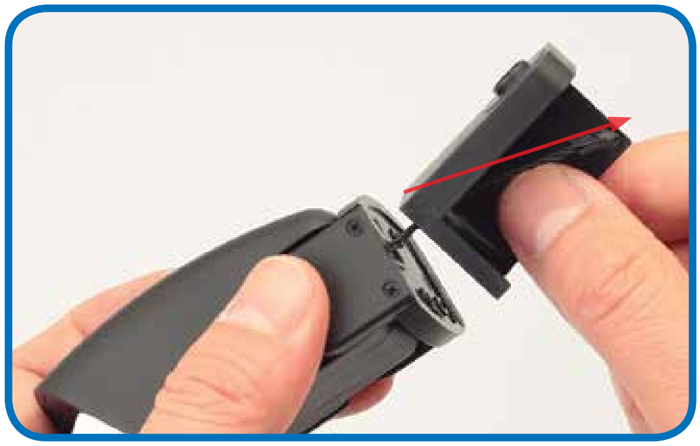
Make absolutely sure that the servo cable is sitting neatly in the recess circled in the photo, and not trapped between the parts (as shown below). If the cable is crushed between the frame and panel, there is a risk that it may short-circuit when in operation, and cause Robi to malfunction. Check that your cable is safe before proceeding.



5 When you are happy that the elbow panel is fitted correctly and the servo cable is not caught in any way, use two M2 x 6mm countersunk screws to secure it.

ADDING THE LEFT UPPER ARM FRAME

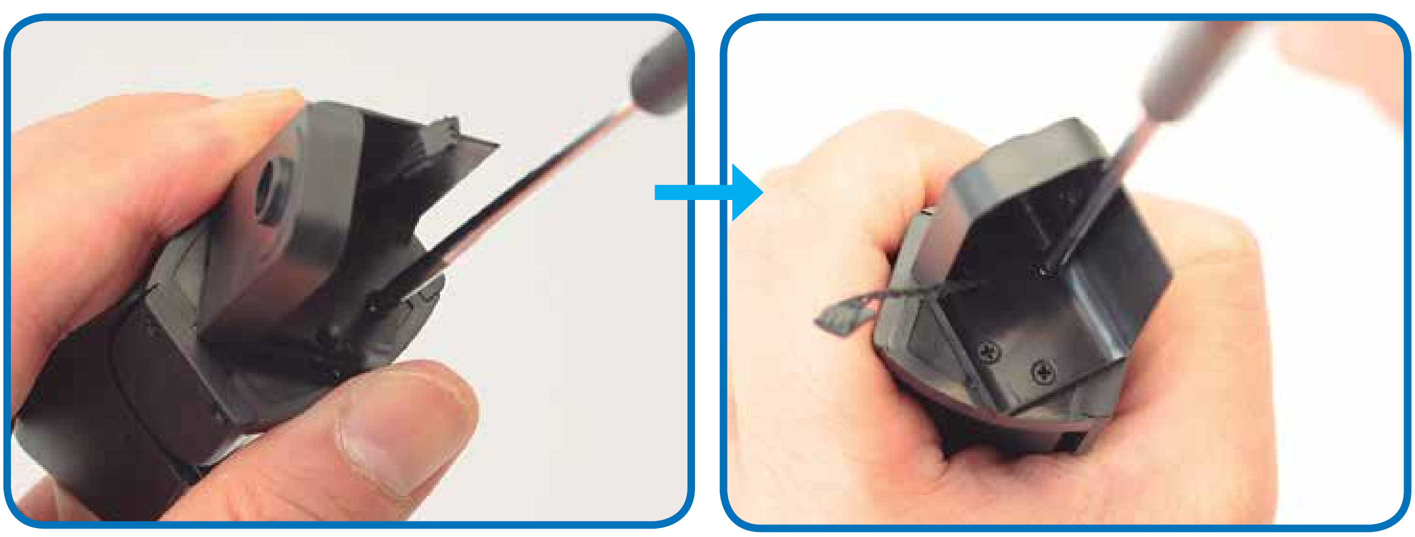
6 Line up the left forearm assembly and the left upper arm frame, noting the position of the rectangular hole (circled).



7 Carefully pass the tip of the servo cable through the rectangular hole, as shown.

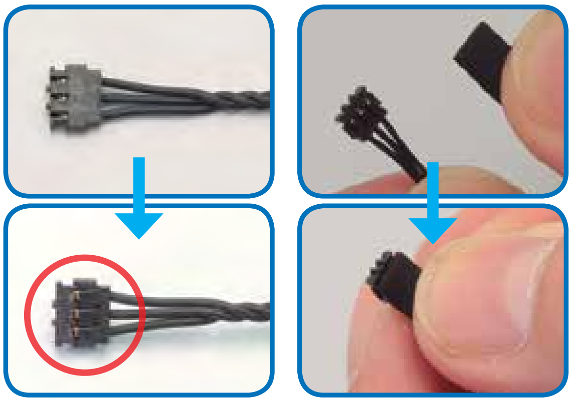
8 With the parts held at a 45° angle to one another, line up the three circled screw holes with those of the forearm beneath.

9 Once you have the screw holes lined up, hold the parts together in one hand, making sure that they do not slip out of position.



10 Still holding the parts in one hand, tighten three more of the M2 x 6mm countersunk screws into the holes lined up in the previous step.

PREPARING THE SERVO CABLE



11 As you have done in previous stages, identify the side of the servo cable's connector marked 'PUSH'. Then, separate a protective pad from the sheet (Stage 3) and place this on top of the connector, making sure that the pad does not extend beyond the connector's tip. Repeat for the connector at the other end of the servo cable so that both are covered.

Assembled left arm and servo cable

Robi's left arm is nearly complete, ready to be fitted with another servo to operate his shoulder movement, and your next servo cable is protected and ready for use in the coming stages.

SERVO CABLES

CLOSE-UP

By now you will be familiar with the servo cables that connect Robi's servo motors, but take a moment to revisit the process of applying the protective pads to these. It is important that the pads are fitted neatly and in line with the shape of the cables' rectangular connectors, because if fitted incorrectly they can obstruct the other cables and components of the servos. For some steps, it will be easier to fit the protective pad after the servo connector is in position – when this is the case, a Tip box will appear next to the step.

The pad is fitted neatly, in line with the connector.

The pad is fitted incorrectly, at an angle to the connector.

COMING IN PACK 6

Your next FOUR complete Stages, as Robi starts to come to life!

Stage 19
Add another servo to Robi's left arm

PARTS PROVIDED
The servo that will operate Robi's left upper arm.

Stage 20
Build up Robi's left shoulder joint

PARTS PROVIDED
Components to build up Robi's shoulder joint, along with another servo cable.

Stage 21
Test Robi's left shoulder servo

PARTS PROVIDED
The servo that will perform the role of Robi's shoulder.

Stage 22
Continue building up Robi's body

PARTS PROVIDED
The next parts of Robi's body, including the left body cover and servo horn.